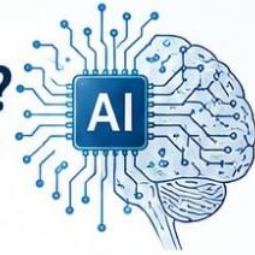


Does Biology Compute Like Artificial Intelligence?

Insights from Living Systems for the Future of Intelligence

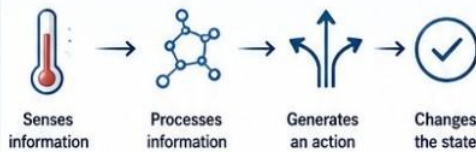
Living cells don't have CPUs or code, yet they sense, decide, adapt, learn and survive. Biology and AI solve similar computational problems—but in profoundly different ways.



1 WHAT IS COMPUTATION?

At its simplest, computation is a transformation:

One state → another state
according to some set of constraints.



Both AI systems and living cells transform information to achieve goals.

2 BIOLOGY vs ARTIFICIAL INTELLIGENCE

Artificial Intelligence

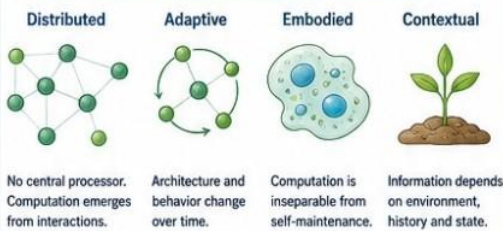
- Engineered architecture
- Fixed hardware
- Trained on data
- Clear objective/loss
- Separated from body and environment
- Can be reset or restarted

VS.

Biological Systems

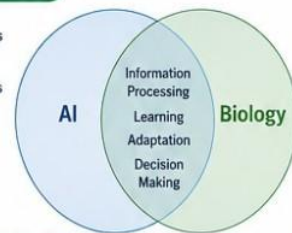
- Self-organizing architecture
- Continuously changing
- Learn while living
- Multiple, context-dependent goals
- Embodied and embedded in environment
- Must survive and maintain itself

3 BIOLOGY COMPUTES DIFFERENTLY



4 WHERE BIOLOGY AND AI MEET

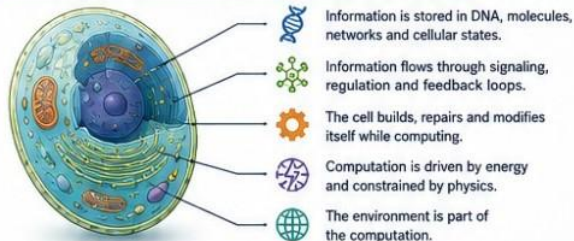
- Pattern Recognition** – Both identify patterns in complex data.
- Representation** – Both build internal models of the world.
- Learning & Adaptation** – Both improve with experience.
- Search & Optimization** – Both explore many possibilities.
- Uncertainty** – Both operate with incomplete information.



5 BIOLOGY TEACHES AI

- Self-Organization** – Build systems that organize themselves.
- Robustness** – Compute reliably despite noise and damage.
- Embodiment** – Intelligence emerges from interaction with the world.
- Continual Learning** – Learn without catastrophic forgetting.
- Energy-Aware Computation** – Use energy as a fundamental resource.

6 THE CELL: AN EXTRAORDINARY COMPUTER



7 A NEW PERSPECTIVE



KEY IDEA

Biology and AI are two different solutions to the same fundamental problem: How can matter process information and generate adaptive behavior?



The cell does not merely run a program. It performs computation while building itself, interacting with the world, and pursuing life.



COMING NEXT

Article 7:
Is Intelligence Older Than Brains?

There is a strange reversal taking place in science.

For decades, biology has borrowed ideas from computing.

We use computers to analyse genomes. We use machine learning to predict protein structures. We use neural networks to recognize patterns in biological data. We use algorithms to reconstruct evolutionary relationships, model cellular networks and search enormous molecular spaces.

It is natural, therefore, to think of the relationship in one direction:

Computers help us understand biology.

But perhaps the more interesting question is the reverse:

Can biology help us understand computation?

What if some of the computational principles that we are now trying to build into artificial intelligence have existed in living systems for billions of years?

This does **not** mean that a cell is secretly running a version of ChatGPT.

A cell does not have a processor, a programming language, a database server or a transformer architecture.

The comparison is more fundamental.

Both artificial intelligence and living systems face a common problem:

How can a physical system transform information into useful action?

Once we ask the question in this way, the similarities—and the differences—become fascinating.

From machines that calculate to systems that adapt

The earliest computers were built to perform calculations.

Give the machine an input, execute a sequence of operations and obtain an output.

The architecture was deliberately designed.

Hardware was separated from software. Memory was separated from processing. Instructions were specified in advance.

Modern artificial intelligence has moved considerably beyond this picture.

A neural network does not need to be explicitly programmed with every rule it will use. Instead, it can learn patterns from examples by modifying internal parameters.

A sufficiently large network can discover relationships that were never explicitly written into its instructions.

Transformers have pushed this idea further.

They can process enormous amounts of information, identify relationships between elements of a sequence, construct internal representations and generate new outputs based on patterns learned from previous data.

Yet even these remarkable systems remain recognizably engineered machines.

They have a designed architecture.

They operate according to mathematical rules chosen by their creators.

They are trained using carefully constructed procedures.

And when something goes wrong, we generally look for the problem in the data, architecture, objective function or training process.

Living systems are different.

A cell does not merely process information.

It continuously changes while processing it.

Its machinery is being built, modified, degraded and rebuilt while the system is operating.

Its environment changes.

Its energy supply changes.

Its internal state changes.

Its molecular components fluctuate.

And yet the cell can often maintain remarkably stable functions.

That raises a deeper question:

What kind of computation can continue while the computer itself is constantly changing?

The cell as a very different kind of computer

Imagine building an artificial computer in which every component can manufacture new components, replace damaged components, alter its own connections and modify its own operating conditions.

Now imagine that the computer must obtain its own energy from the environment.

It must repair itself.

It must reproduce.

It must respond to threats.

It must change its internal configuration according to circumstances.

And it must do all this without a central processor.

That begins to sound less like a conventional computer and more like a cell.

The comparison is not merely philosophical.

In my earlier work, I proposed that biological networks can be viewed as information-processing systems and explored the possibility that they may employ both classical and quantum forms of information processing.

I also explored the possibility that interactions between biological molecules could be interpreted computationally, including the more speculative hypothesis that molecular systems might exploit quantum states. These ideas remain hypotheses rather than established biological principles.

The important point for the present discussion is broader:

Biology does not need to resemble a silicon computer in order to compute.

Perhaps we have been defining computation too narrowly.

What does it actually mean to compute?

The word *computation* can sound more mysterious than it really is.

At its simplest, computation means a transformation:

one state → another state

according to some set of constraints.

A calculator transforms numbers.

A neural network transforms patterns.

A thermostat transforms temperature information into a control action.

A biological signalling pathway transforms molecular states into another set of molecular states.

A cell receiving a signal from its environment may change gene expression, metabolism, movement, growth or survival.

In each case, something has changed because information about the state of the system has been transformed.

This does not mean that every chemical reaction is automatically an intelligent computation.

That would make the word *computation* almost meaningless.

The interesting question is whether biological systems contain **organized networks of state transformations that collectively perform functions such as sensing, integration, memory, decision-making, prediction or adaptation.**

And increasingly, biology gives us many reasons to ask that question.

Neural networks are not brains

The comparison with artificial neural networks is particularly revealing.

An artificial neural network contains units connected by adjustable parameters.

During training, those parameters are modified so that the network becomes better at performing a task.

The network gradually acquires an internal representation of patterns in its training data.

Biological systems also contain enormous networks of interacting components.

Genes regulate genes.

Proteins regulate proteins.

Metabolites influence enzymes.

Signalling molecules influence transcription.

Membranes regulate molecular traffic.

Cells communicate with neighbouring cells.

Tissues communicate with one another.

At every level, states influence other states.

But there is a crucial difference.

In an artificial neural network, the architecture is generally specified by the engineer before training begins.

In biology, **architecture itself is part of the problem being solved.**

The organism must develop its architecture.

A fertilized egg does not contain a miniature adult waiting to unfold.

It contains a developmental system that generates an organism through a dynamic interaction among genetic information, molecular interactions, physical constraints and environmental conditions.

This makes biological computation fundamentally interesting.

The system is not simply learning **within** an architecture.

It is also participating in the formation and modification of its architecture.

Evolution is another kind of learning

There is an even more remarkable comparison.

Artificial intelligence systems can learn during their lifetime.

Biological populations can change across generations.

Evolution is therefore sometimes described, at a very broad conceptual level, as a learning process.

But evolution is not simply biological gradient descent.

There is no single loss function written on a whiteboard telling evolution exactly what to optimize.

There is variation.

There is inheritance.

There is interaction with the environment.

There is differential survival and reproduction.

There are developmental constraints.

There are historical contingencies.

And there are enormous numbers of parallel experiments occurring across populations and generations.

The resulting system can display astonishing adaptation without any central designer.

This raises an intriguing computational question:

Could natural selection be understood partly as an enormous distributed search process?

The analogy should not be pushed too far.

Evolution does not necessarily "know" the solution it is searching for.

It does not operate like an engineer running a standard optimization algorithm.

But both evolution and machine learning confront a similar abstract problem:

How can a system move through an enormous space of possibilities and retain configurations that work better under particular conditions?

This perspective may help us see evolution not simply as a historical process, but also as a process that progressively shapes the information-processing architecture of living systems.

But biology has something AI does not

There is an important asymmetry in the comparison.

Artificial intelligence systems are usually built to perform a task.

A language model is trained to predict patterns in language.

An image model is trained to recognize or generate images.

A reinforcement-learning agent may be trained to maximize a reward.

Biological systems have a much more difficult objective.

They must remain in existence.

This means that biological computation is embedded in **self-maintenance**.

The system must regulate itself while simultaneously interacting with its environment.

A cell must maintain concentrations, energy balance, membrane integrity, molecular organization and many other variables.

This gives biological information processing a character that is different from ordinary artificial computation.

The computer can fail and be restarted.

The cell cannot simply reboot.

Its computation is intimately connected to its continued existence.

The computer changes. The program changes. The goal changes.

This may be the most important difference of all.

Imagine an AI system that changes its own hardware while learning.

Its neural connections change.

Its memory architecture changes.

Its sensors change.

Its energy budget changes.

Its objective can change.

And yet it continues operating.

That sounds extraordinarily difficult to engineer.

For biological systems, it is normal.

During development, cells divide and differentiate.

Networks form.

Connections disappear.

Gene expression changes.

Proteins are synthesized and degraded.

Cells respond to mechanical forces.

Tissues reorganize.

An organism ages and repairs itself.

In other words:

The biological computer is not separate from the process being computed.

The computation and the construction of the computing system are intertwined.

This is one reason why biological intelligence may ultimately teach computer science something that conventional AI has largely ignored:

how to compute while continuously rebuilding the machine that performs the computation.

What about transformers?

The comparison becomes even more interesting when we look at transformers.

Transformers became revolutionary because they provided an efficient way of modelling relationships among elements of a sequence.

The mechanism commonly called *attention* allows the system to weight relationships between different parts of its input.

A biological system obviously does not contain a transformer architecture in the engineering sense.

But biology is filled with systems in which the meaning of a component depends on its relationships with other components.

A gene does not operate in isolation.

Its effect depends on regulatory networks.

A protein's function depends on its partners, location, modification state and cellular context.

A metabolite's significance depends on the metabolic state of the cell.

A cell's behaviour depends partly on signals from neighbouring cells.

This suggests a general principle:

Information in biology is relational.

The state of one component cannot always be understood independently of the network in which it participates.

This is very close to the conceptual shift from thinking about isolated symbols to thinking about relationships among representations.

But again, the analogy should not be mistaken for identity.

A biological regulatory network is not literally a transformer.

The point is that **similar computational problems can arise in radically different physical architectures.**

Biology does not have a single processor

A conventional computer has a processor.

Many AI systems have clearly defined computational architectures.

But where is the processor of a cell?

There isn't one.

Computation is distributed.

A molecular interaction changes another molecular state.

That state influences another reaction.

Several pathways converge.

Feedback loops regulate activity.

Signals propagate through networks.

The resulting cellular state emerges from the collective dynamics.

This is remarkably different from the idea of a central executive.

In fact, one of the recurring themes in my earlier work has been that biological networks may possess computational properties that cannot be adequately represented simply by examining their individual components. My work on logical possibilities in biological networks similarly explored how collective behaviour can produce outcomes that are not obvious from individual molecular decisions.

This suggests another possible lesson for AI:

Intelligence may not require a central place where "the computation" happens.

It may emerge from interactions among many simpler processes.

The mystery of collective computation

Consider a simple hypothetical cellular network.

Suppose three molecular pathways respond differently to the same environmental condition.

One favours one state.

Another favours a second state.

A third favours an intermediate state.

If we examined each pathway independently, each could appear perfectly reasonable.

But when the pathways interact, the collective behaviour can become surprisingly complicated.

In my earlier work on logical behaviour in biological networks, I explored examples where collective molecular preferences can produce intransitive or paradoxical outcomes analogous to problems studied in collective decision theory.

The significance is not that cells are holding elections.

They obviously are not.

The deeper point is that:

the logic of a network is not necessarily the sum of the logic of its components.

This is an important lesson for artificial intelligence.

If we understand each artificial neuron perfectly, do we therefore understand the intelligence of the entire network?

Not necessarily.

Just as understanding every transistor in a computer does not by itself explain the meaning of the software running on it, understanding every molecular interaction may not fully capture the system-level behaviour of a living organism.

Where does memory live?

Here the comparison becomes even more difficult.

In an AI model, learned information is represented partly in its parameters.

Change the parameters sufficiently, and the learned behaviour changes.

Biology has DNA.

It is tempting to say:

DNA = biological memory.

But that is only part of the story.

DNA contains information, but the meaning and effect of that information depend on cellular context.

The same genome can participate in very different cellular states.

Cells differentiate.

Genes are switched on and off.

Proteins modify one another.

Metabolic conditions alter activity.

Environmental signals feed back into the system.

So perhaps the biological equivalent of memory is distributed across multiple levels.

Some information is encoded in DNA.

Some is represented by molecular concentrations.

Some exists in regulatory states.

Some is contained in cellular architecture.

Some may exist in persistent network states.

And some information exists at the level of the organism's interaction with its environment.

This is quite unlike the simple idea of a single biological "hard drive."

A thought experiment

Imagine that we take two cells with identical DNA.

We place them in different environments.

One receives nutrient A.

The other receives nutrient B.

The cells begin to behave differently.

Their molecular concentrations change.

Different genes become active.

Different proteins are produced.

Different metabolic pathways dominate.

Eventually, the two cells may occupy very different physiological states.

Where did the difference come from?

Not from a difference in their DNA sequence.

The difference emerged from the interaction between:

genome + cellular state + environment + history.

Now imagine that we somehow return both cells to the same external environment.

They may not immediately become identical.

Their internal states retain some history.

This gives us a powerful conceptual possibility:

Biological memory may not be stored in one location. It may be distributed across the state of the system.

This is a very different architecture from the conventional computer.

Development: computation that builds a body

Perhaps the strongest challenge to conventional AI comes from development.

Consider an embryo.

Starting from a single fertilized cell, a complex organism emerges.

Cells divide.

They communicate.

They differentiate.

They migrate.

They respond to physical forces.

They change their gene expression.

They interact with neighbouring cells.

Structures form.

Errors are corrected.

The organism eventually reaches a recognizable anatomical organization.

This is not simply information retrieval.

It resembles **distributed problem-solving**.

And the problem is extraordinary:

How can a collection of cells construct a coherent organism without a central engineer supervising every step?

Developmental biology suggests that the answer involves multiple interacting information systems rather than a single linear genetic program.

Genes matter enormously.

But genes operate within a physical and cellular environment.

The developmental system therefore looks less like a computer executing a fixed program and more like a **self-organizing computational process**.

The genome is not the whole algorithm

This point deserves emphasis.

If I give you the source code of a computer program, you may be able to reproduce its behaviour.

But if I give you a genome, I have not necessarily given you everything required to reconstruct the organism.

You also need the cellular environment.

You need molecular machinery.

You need developmental history.

You need physical conditions.

You need interactions among cells.

You need environmental information.

And, ultimately, you need the historical process through which the organism evolved.

The genome is therefore better thought of as one component of a much larger information-processing system.

This does not diminish DNA.

Quite the opposite.

It places DNA within the larger architecture of biological information.

AI learns from examples. Biology learns from existence.

There is another profound difference.

An AI system can be trained on millions of examples without ever being in danger of dying.

A biological organism cannot separate learning completely from survival.

Its sensing, decision-making, metabolism and reproduction are intertwined.

A bacterium moving toward nutrients is not solving an abstract classification problem.

Its information processing is connected directly to whether it survives.

A plant adjusting its growth in response to light is not merely recognizing a pattern.

Its response changes its future.

An immune system identifying a pathogen is not simply making a prediction.

It is initiating a coordinated biological response.

This suggests that biological intelligence is deeply **embodied**.

Information is not processed for its own sake.

It is processed because the organism is a physical system trying to remain viable in a changing environment.

Could AI become more biological?

Perhaps the comparison should therefore be reversed.

Instead of asking:

How can we make biology more like AI?

we might ask:

How can we make AI more like biology?

That would mean developing systems that are:

self-organizing,

adaptive,

distributed,

energy-aware,

robust to damage,

capable of modifying their own architecture,

embedded in an environment,

and capable of pursuing persistent goals under changing conditions.

Some areas of AI and robotics are already moving in these directions.

Evolutionary computation explores search inspired by evolution.

Developmental robotics explores how intelligent behaviour can emerge through development and interaction.

Neuromorphic computing attempts to borrow principles from nervous systems.

Active learning allows systems to decide which information is worth acquiring.

Continual learning tries to prevent the catastrophic loss of previous knowledge when systems learn new tasks.

These are not identical to biology.

But they point toward a future in which AI may increasingly borrow not only **algorithms** from nature, but **architectural principles**.

And perhaps biology can learn from AI

The flow of ideas can also go in the other direction.

AI provides a powerful language for asking new questions about biological systems.

What is the representation?

What is the objective?

What is the memory?

Where is the feedback?

What constitutes learning?

What is the system optimizing?

How is information distributed?

How does the system deal with uncertainty?

How does it explore alternatives?

These questions can be asked at the level of molecules, cells, tissues, organisms and populations.

This could eventually lead to a different kind of computational biology.

Not simply:

using computers to analyse biological data

but:

using computation as a conceptual language for understanding biological organization.

That is a much larger ambition.

But is biology actually "intelligent"?

Here we need to be careful.

If we call every adaptive system intelligent, the word eventually loses its meaning.

A thermostat responds to temperature.

A chemical reaction responds to concentrations.

A crystal self-organizes.

A cell responds to its environment.

A human being reasons about the future.

These are clearly not equivalent.

So we need levels.

Perhaps there is a continuum:

physical response → regulation → information processing → adaptation → learning → problem-solving → intelligence → cognition

The boundaries are not necessarily sharp.

And this is precisely where the next stage of the *Rethinking Biology* series will take us.

If computation can exist without a brain, perhaps intelligence can also exist without a brain.

That is a much more provocative proposition.

The deeper question: what is computation for?

Artificial intelligence has forced us to confront a question that biology has been asking for billions of years:

What should a system do when it does not know what will happen next?

A chess-playing AI searches possible moves.

A language model generates possible continuations.

An autonomous robot searches possible actions.

A biological organism also faces an uncertain future.

It must decide where to move.

What to eat.

When to grow.

When to reproduce.

How to respond to danger.

How to repair itself.

How to allocate limited resources.

But biological systems do not perform these tasks through a single centralized search algorithm.

They distribute the problem across multiple levels of organization.

Molecules respond.

Cells integrate.

Tissues coordinate.

Organs specialize.

Organisms act.

Populations evolve.

The result is an enormous hierarchy of interacting information-processing systems.

Perhaps this is one of the great computational discoveries of evolution:

You do not necessarily need one very powerful computer if you can build a society of many imperfect computers that cooperate.

From neural networks to living networks

This may eventually change how we think about artificial intelligence.

The dominant metaphor today is still largely that of the artificial neural network.

But perhaps the next generation of AI will look less like a neural network and more like an ecosystem.

Instead of one model, there may be many interacting subsystems.

Instead of a fixed architecture, there may be continual restructuring.

Instead of one objective, there may be competing objectives.

Instead of centralized training, there may be local adaptation.

Instead of a static dataset, there may be continuous interaction with the environment.

Instead of computation being separated from maintenance, the system may maintain itself while computing.

Such systems could be much closer to biological organization.

But they would also be much harder to understand.

And that brings us back to biology.

The biological computer may be unlike anything we have built

The more we compare biology with artificial intelligence, the less obvious the analogy becomes.

A biological system does not look like a computer because it was not designed to look like one.

It evolved.

Its architecture is historical.

Its computation is embodied.

Its memory is distributed.

Its hardware is plastic.

Its software is inseparable from its hardware.

Its energy flows through its computation.

Its environment participates in the process.

Its components communicate at many scales.

And the system itself is continually changing.

This may explain why biological systems can be extraordinarily robust despite being noisy and imperfect.

A conventional computer is often designed to eliminate noise.

Biology appears to have learned how to **compute despite noise**.

A conventional computer tries to preserve its architecture.

Biology continually **rebuilds its architecture**.

A conventional computer separates program from machine.

Biology **integrates program, machine and environment**.

And perhaps most remarkably:

A conventional computer is built to serve a purpose specified by someone else.

A biological system has evolved mechanisms that allow it to maintain itself and pursue its own persistence.

So, does biology compute like AI?

The answer is both **yes and no**.

Yes, in the broad sense that both biological and artificial systems can transform information, integrate signals, recognize patterns, adapt internal states and generate actions.

No, in the sense that the underlying architectures are profoundly different.

Biology is not an artificial neural network.

A cell is not a transformer.

Evolution is not simply gradient descent.

DNA is not simply software.

And the organism is not simply a machine executing a program.

The more interesting conclusion is therefore not:

"Biology is an AI."

It is:

"AI and biology may be two very different solutions to the same fundamental problem: how can matter process information and generate adaptive behaviour?"

That is a much more useful question.

From AI back to biology

This brings us to a remarkable possibility.

Perhaps we have entered a period in which biology and artificial intelligence can begin to illuminate one another.

Biology can provide AI with ideas about:

self-organization, embodiment, distributed computation, adaptation, development, resilience and continual learning.

AI can provide biology with ideas about:

representation, computation, search, learning, optimization, memory and information flow.

The two fields may therefore meet somewhere in the middle.

Not because biology is secretly a computer.

And not because computers are becoming biological.

But because both are revealing something deeper about information.

Information is not merely something that computers manipulate.

It may be something that organized matter itself can use to maintain structure, respond to uncertainty and navigate the future.

That takes us one step further in the journey.

We began this series by asking whether biology needs to be understood partly as an information science.

We then asked whether the cell is more than a collection of chemicals.

We explored quantum biology.

We asked whether cells might perform computation.

We examined the possibility that molecular interactions themselves could participate in information processing.

Now we have reached a new question.

If computation can exist in living matter without a conventional computer, can intelligence exist without a conventional brain?

A bacterium has no brain.

A plant has no brain.

An immune system has no brain.

A slime mould has no brain.

And yet these systems can sense, respond, adapt and solve problems.

Perhaps intelligence did not begin when brains appeared.

Perhaps brains represent one particularly powerful solution to a much older biological problem.

The problem of making useful decisions in an uncertain world.

That is where we go next.

The next question

Article 7 — Is Intelligence Older Than Brains?

If information processing existed long before nervous systems, perhaps intelligence did too.

The next article will explore plants, bacteria, immune systems, slime moulds and collective biological behaviour—and ask whether the history of intelligence may be much older, and much broader, than the history of brains.

Because perhaps the real question is not:

"When did animals become intelligent?"

but:

"When did matter first begin to solve problems?"

Rethinking Biology: Essays on Information, Intelligence and Life

Article 6 of 12

The series is deliberately moving from established science toward increasingly speculative questions. Articles 1–4 establish the scientific foundation; Articles 5–8 explore scientific extrapolations; later articles will introduce more explicitly theoretical ideas.

And one principle will remain important throughout:

A hypothesis is most useful when we can ask how it might be wrong.

That is the standard I intend to apply as this journey continues.